Philosophy of science for linguists
1. Scientific explanation

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Doctoral seminar: Linguistic Argumentation
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Opening discussion

Is linguistics a science?

Why philosophy of science for linguists?

'philosophy of science is about as useful to scientists
as ornithology is to birds'
(Richard Feynman – perhaps apocryphal)

However:

- Appeal is made to these notions in theorizing about what good linguistics looks like
- Disputes between different schools of thought in linguistics may hinge on what they consider to be good linguistics
- In any case: Explicit thinking about different forms of reasoning is useful when constructing arguments



Frame from Look Around You series 1 episode 2, 'Water'

What is science?

- The idea of science was certainly familiar to the Greeks 2,500 years ago (Rosenberg, 2005, 13)
- The English term (from Old French, from Latin) is first attested less than 1,000 years ago, in the Middle English period
- The word *scientist* was coined by William Whewell in 1834 to describe Scottish polymath Mary Somerville $\longrightarrow$



- In the 18th and 19th centuries, *science* had roughly the same reference as German *Wissenschaft*, i.e. academic knowledge/disciplines
- In the 20th century, in English, the term came to refer primarily to the natural sciences (German *Naturwissenschaften*)
- Let's sidestep the question and return to scientific explanation later

What is philosophy?

- Rosenberg (2005, 4): 'Philosophy deals with two sets of questions:
  1. First, the questions that science – physical, biological, social, behavioral – cannot answer now and perhaps may never be able to answer.
  2. Second, the questions about why the sciences cannot answer the first lot of questions.'
- Philosophy includes:
  - **Logic:** the study of reasoning
  - **Epistemology:** the study of knowledge
  - **Metaphysics:** the study of the fundamental structure of reality

Descriptivism and prescriptivism in science

- Linguistics 101: the discipline is **descriptive**, not prescriptive (**normative**).
- Science in general is descriptive.
- Philosophy may be (but need not be).
  - 'the sciences – social or natural – do not challenge or defend the normative views we may hold. This is a task for philosophy' (Rosenberg, 2005, 4)
- BUT: aspiring scientists – and linguists – sometimes turn to philosophy of science for normative rules about how to achieve scientific explanation (Rosenberg, 2005, 22)

Today's starting point: empiricism

Philosophical position, dominant through most of the twentieth century.

- Key idea: **knowledge is justified by experience** (building on Locke, Berkeley, Hume in 17th century)
- Empiricism holds scientific truths to be contingent, not necessary
- Maths is a problem for empiricism: mathematical truths are necessary
  - e.g. $2 + 2 = 4$

Contrasts with **rationalism**, the idea that knowledge is justified primarily by reason (building on Descartes, Spinoza, Leibniz in 17th century)

Logic refresher: deductive vs. inductive reasoning

Deductive inferences are always valid, if the premises are true.

1. Definite articles in German have case.
2. *dem* is a definite article in German.
3. ∴ *dem* has case.

Inductive inferences are fallible.
E.g. for English:

1. The plural of *house* is *houses*.
2. The plural of *dog* is *dogs*.
3. The plural of *cheese* is *cheeses*.
4. ...
5. ∴? Plurals are formed with *-s*.

Science is faced with the 'problem of induction' (Hume).

Logical positivism



Meeting place of the Vienna Circle (photo by Institute Vienna Circle, CC-BY-SA 4.0)

- Developed in Europe between the wars (Vienna Circle)
- A flavour of empiricism, combined with mathematical logic
- **Principle of verifiability:** 'every meaningful (i.e. true or false) statement about the world is one that can be verified (or at least tested) by experience' (Rosenberg, 2005, 23)
- Non-scientific fields (maths, philosophy) are really just about the consequences of definitions – tautologies
- Strongly prescriptive/normative

Key questions in philosophy of science

Logical positivists set the agenda.

- What is an explanation?
- What is a scientific law?
- What is a theory?
- What is the role of empirical evidence in deciding between competing hypotheses?
- How do we choose between hypotheses in the absence of evidence?

Underdetermination of theories by evidence

Example: /l/ in English has two allophones.

- **Light l:** [l] in the onset of a syllable, e.g. **l***ip*.
- **Dark l:** [ɬ] in the coda of a syllable, e.g. *mi***ll**.

Compatible rule: /l/ $\longrightarrow$ [ɬ] / _[+coda].

But also:

- /l/ $\longrightarrow$ /r/ $\longrightarrow$ [ɬ] / _[+coda]
- /l/ $\longrightarrow$ /p/ $\longrightarrow$ [ɬ] / _[+coda]
- /l/ $\longrightarrow$ /p/ $\longrightarrow$ /r/ $\longrightarrow$ [ɬ] / _[+coda]

How do we decide? (On this problem see Rosenberg, 2005, 138–140.)

What is scientific explanation?

- Widespread view: good scientific explanations should contain laws
- We're looking for **necessary** and **sufficient** conditions for a scientific explanation
  - A **necessary** condition must be present in order for another condition to occur
    - Being sittable-on is a necessary condition for being a *chair*
    - Having language is not a necessary condition for being a *human*
  - If a **sufficient** condition is present, then another condition must occur
    - Being sittable-on is not a sufficient condition for being a *chair*
    - Having language is a sufficient condition for being a *human*
- Note: Scientific explanation is not necessarily causal (see Rosenberg, 2005, 27–28)

Deductive-nomological explanation

Based on the work of Carl G. Hempel (e.g. Hempel & Oppenheim, 1948):

1. The explanation must be a valid deductive argument.
2. The explanation must contain at least one general law actually needed in the deduction.
3. The explanation must be empirically testable.
4. The sentences in the explanation must be true.

For discussion in the context of language change, see Lass (1980).

Example 1: Neogrammarian sound laws

The Neogrammarians: scholars, mainly in Germany, in the nineteenth century who posited exceptionless ('regular') sound changes.

1. Grimm's Law in Germanic: /p/ $\longrightarrow$ /f/.
2. The word *pisk- for 'fish' is inherited into Germanic.
3. $\therefore$ the Germanic word for 'fish' becomes *fisk-.

- Are Neogrammarian sound laws 'laws' in the scientific sense? Why (or why not)?
- Are they causal explanations?

Example 2: Verb-second

1. Finite verbs may occupy second position in German.
2. 'V2 Law' (after den Besten, 1989): complementizers block finite verbs from moving to the C position.
3. ∴ the verb is in second position in main clauses, but not in (most) embedded clauses.

- Is this a 'law' in the scientific sense? Why (or why not)?
- Is it a causal explanation?

Explanation in linguistics

Explanation in linguistics has been a hotly contested notion.

- 'To the extent that a linguistic theory succeeds in selecting a descriptively adequate grammar on the basis of primary linguistic data, we can say that it meets the condition of *explanatory adequacy*. That is, to this extent, it offers an explanation for the intuition of the native speaker on the basis of an empirical hypothesis concerning the innate predisposition of the child to develop a certain kind of theory to deal with the evidence presented to him.' (Chomsky, 1965, 29–30)
- 'In essence, a formal model is *nothing* but a restatement of the facts at a tighter level of generalization ... There is one thing, however, that a formal model can never do: It cannot *explain* a single thing. ... The history of transformational-generative linguistics boils down to nothing but a blatant attempt to represent the formalism as 'theory', to assert that it 'predicts a range of facts', that it 'makes empirical claims', and that it somehow 'explains'.' (Givón, 1979, 5–6)

Problems for deductive-nomological explanation

Counterexamples to the idea of deductive-nomological (D-N) explanation:

1. Explanations which everyone agrees to be an explanation but which fail to fit the D-N model.
   - e.g. Britain entering WWI against Germany – what law is involved?
   - Neogrammarian sound laws: not general
2. Explanations which fit the D-N model but aren't explanations (Bromberger, 1966):
   i. Light travels in straight lines. (a law)
   ii. At 3.00p.m. on the 4th of July 2000 the Sun is emitting light at a 45 degree angle to the ground where the flagpole is located, perpendicular to the ground. (boundary condition)
   iii. The shadow cast by the flagpole is 50 feet long. (boundary condition)
   iv. A triangle with two equal angles is isosceles. (mathematical truth)
   v. ∴ The flagpole is 50 feet high.

Pragmatics?

Linguistics enters the picture unexpectedly:

Is explanation a matter of pragmatics?

van Fraassen (1980): explanations must be **relevant** in order to be accepted.

Inductive-statistical explanation

One might think that statistics solves these problems. E.g. in sociophonetics (made-up example):

1. 80% of female speakers 25–39 produce /æ/ as [ɛə]. (well-confirmed generalization)
2. Julie is a female speaker aged 29.
3. ∴ (with 80% probability) Julie's next /æ/ will be produced as [ɛə].

This is not a deductive argument (obviously). Further facts can change our view: for instance, what about the phonological environment?

For more on this, see Rosenberg (2005, 38–44).

Big Data

Does 'Big Data' constitute a whole new way of doing science?

Leonelli (2014): not really.

- Large amounts of data are not themselves new
- What's new is:
  1. 'the prominence and status acquired by data as scientific commodity and recognised output both within and beyond the sciences'
  2. 'the methods, infrastructures, technologies and skills developed to handle (format, disseminate, retrieve, model and interpret) data' (Leonelli, 2014, 2)
- More focus on 'data journeys' needed
- The transformative role of Big Data may be more important in the social and behavioural sciences – for better or for worse

Explanation outside the 'hard' sciences

'Those who appreciate the power and the successes of the natural sciences, and who wish to apply methods successful in these disciplines to the social and behavioral sciences, have a special incentive to analyze the methods that have enabled natural science to attain its successes. Since the emergence of the social and behavioral sciences as self-consciously 'scientific' enterprises, social and behavioral scientists, and some philosophers of science, have held that the relative lack of success of these disciplines, by contrast to the natural sciences, is due to a failure correctly to identify or implement methods which have succeeded in natural science. For these students of social science, the philosophy of science has an obviously prescriptive role. Once it reveals the features of evidence gathering, the explanatory strategies, and the ways in which both are applied in the natural sciences, the key to similar advance in the social and behavioral science becomes available. All the social and behavioral sciences need to do is employ the right method. Or so these students of scientific methodology argue.' (Rosenberg, 2005, 6)

What do you think? Why?

Summary

- The dominant epistemology in science over the past hundred years (and more) has been a form of **empiricism**.
  - An alternative, influential in philosophy and linguistics: **rationalism**.
- The logical positivist characterization of explanation is problematic.
  - Perhaps what counts as a good explanation is relevance-based.
- Neither inductive-statistical approaches in general nor Big Data in particular solve the problems facing logical positivism, or empiricism in general.
- Open question: is there a meaningful difference in the nature of explanation between the 'hard'/natural sciences and the social and behavioural sciences? Where does linguistics fit?

Readings and next time

- Reading for this session: Rosenberg (2005, chapters 1–3), esp. 2.
- Next session (not next week): **Scientific theories and paradigms**.

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